

1. General Description

The EM74HC573; EM74HCT573 is an 8-bit D-type transparent latch with 3-state outputs. The device features latch enable (LE) and output enable ($\overline{OE}$) inputs. When LE is HIGH, data at the inputs enter the latches. In this condition the latches are transparent, a latch output will change each time its corresponding D-input changes. When LE is LOW the latches store the information that was present at the inputs a set-up time preceding the HIGH-to-LOW transition of LE. A HIGH on $\overline{OE}$ causes the outputs to assume a high-impedance OFF-state. Operation of the $\overline{OE}$ input does not affect the state of the latches. Inputs include clamp diodes. This enables the use of current limiting resistors to interface inputs to voltages in excess of V_{CC} .

2. Features and Benefits

- Wide operating voltage 2.0 V to 6.0 V
- High noise immunity
- CMOS low power dissipation
- Latch-up performance exceeds 250 mA
- Input levels:
 - For EM74HC573: CMOS level
 - For EM74HCT573: TTL level
- Inputs and outputs on opposite sides of package allowing easy interface with microprocessors
- Useful as input or output port for microprocessors and microcomputers
- 3-state non-inverting outputs for bus-oriented applications
- Common 3-state output enable input
- Complies with JEDEC standards:
 - JESD8C (2.7 V to 3.6 V)
 - JESD7A (2.0 V to 6.0 V)
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 2 exceeds 3500 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options
- Specified from -40 °C to +85 °C and from -40 °C to +125 °C

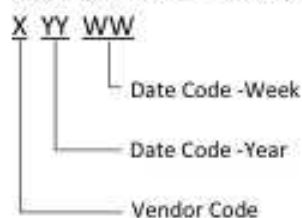
3. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EM74HC573D	HC573 XYYWW	SOP-20L	plastic small outline package; 20 leads; body width 7.5 mm	2000
EM74HCT573D	HCT573 XYYWW			
EM74HC573PW	HC573 XYYWW	TSSOP20L	plastic thin shrink small outline package; 20 leads; body width 4.4 mm	3000
EM74HCT573PW	HCT573 XYYWW			

MARKING INFORMATION

NOTE: XYYWW = Vendor Code and Date Code.



4. Function Diagram

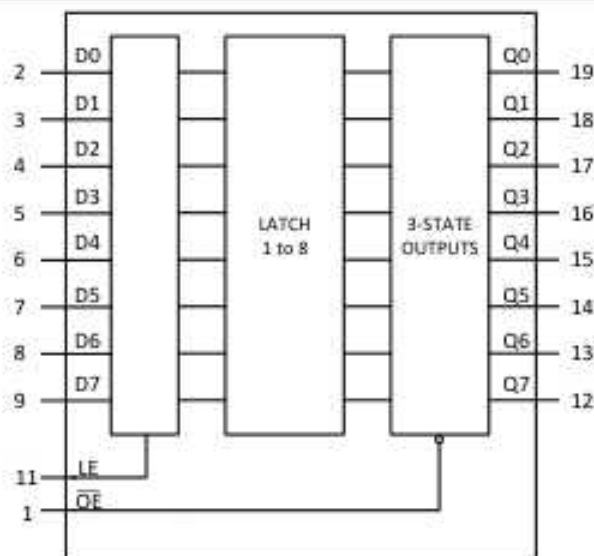


Fig. 1. Logic diagram

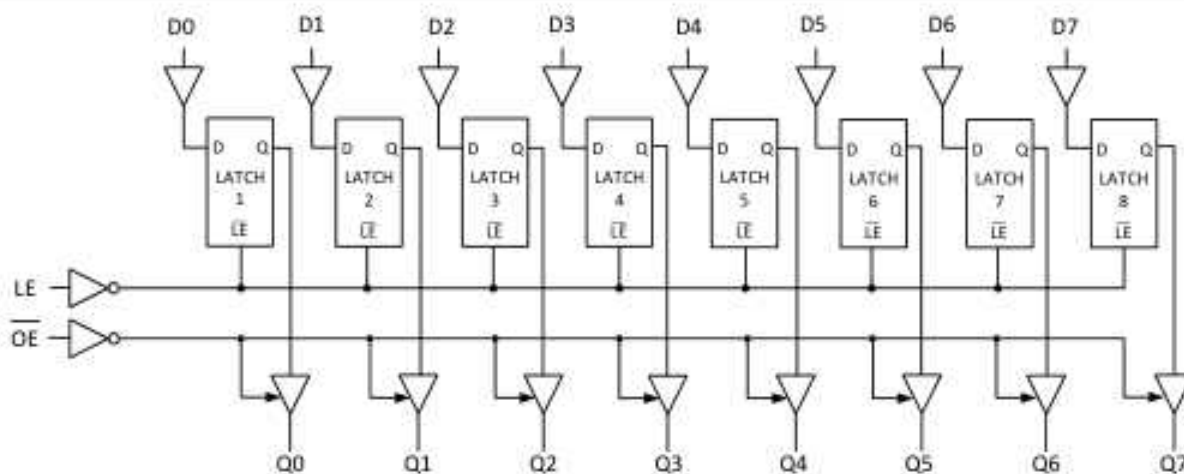
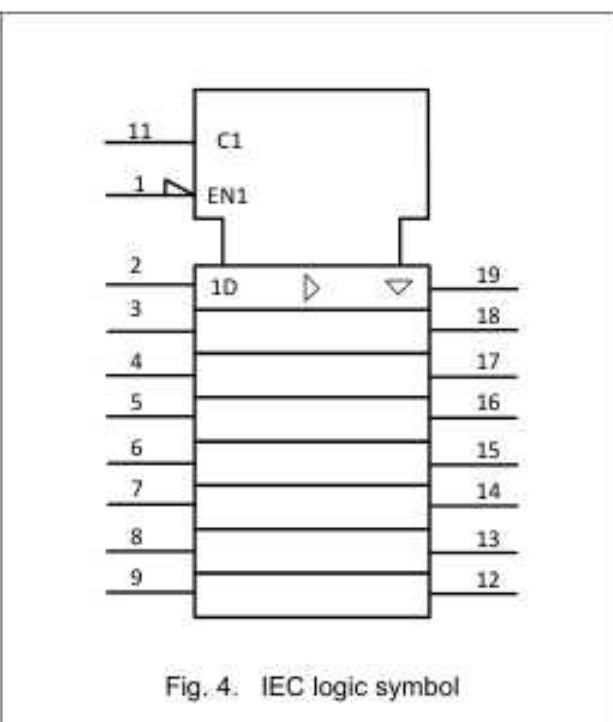
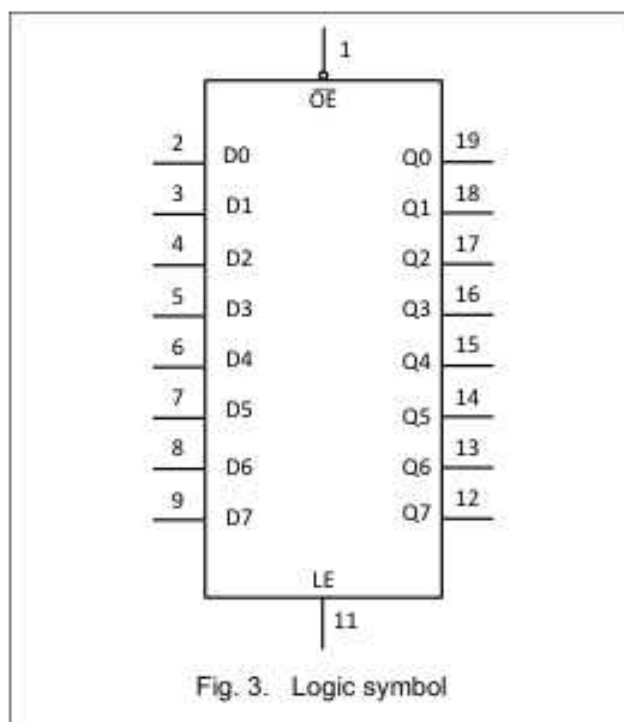


Fig. 2. Functional diagram



5. Pinning Information

5.1. Pinning

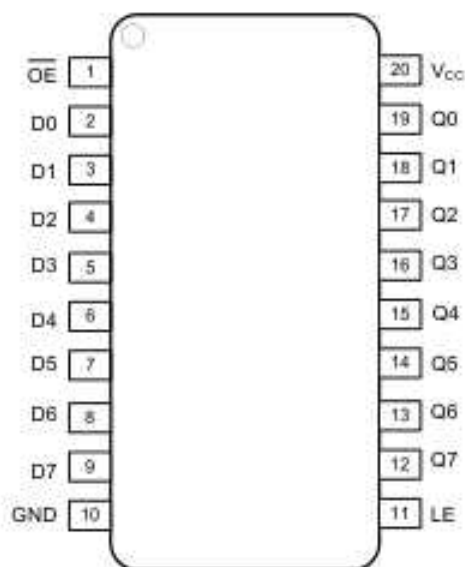


Fig. 5. Top view pin configuration SOP-20L and TSSOP-20L

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5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
$\overline{OE}$	1	3-state output enable input (active LOW)
Q0, Q1, Q2, Q3, Q4, Q5, Q6, Q7	19,18,17,16,15,14,13,12	3-state latch output
D0, D1, D2, D3, D4, D5, D6, D7	2, 3, 4, 5, 6, 7, 8, 9	data input
GND	10	ground (0 V)
LE	11	latch enable input (active HIGH)
V _{CC}	20	supply voltage

6. Functional Description

Table 3. Function table

H = HIGH voltage level; h = HIGH voltage level one set-up time prior to the LOW-to-HIGH clock transition;
 L = LOW voltage level; l = LOW voltage level one set-up time prior to the LOW-to-HIGH clock transition;
 Z = high-impedance OFF-state.

Operating modes	Control		Input	Internal latches	Outputs
	$\overline{OE}$	LE	Dn		Qn
Enable and read register (transparent mode)	L	H	L	L	L
			H	H	H
Latch and read register	L	L	l	L	L
			h	H	H
Latch register and disable outputs	H	L	l	L	Z
			h	H	Z

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 5. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	7	V
I_{IK}	input clamping current	$V_I < -0.5\text{ V}$ or $V_I > V_{CC} + 0.5\text{ V}$		± 20	mA
I_{OK}	output clamping current	$V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$		± 20	mA
I_O	output current	$-0.5\text{ V} < V_O < V_{CC} + 0.5\text{ V}$		± 35	mA
I_{CC}	supply current			70	mA
I_{GND}	ground current		-70		mA
P_{tot}	total power dissipation	$T_{amb} = -40\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$		500	mW
T_{stg}	storage temperature		-65	+150	$^{\circ}\text{C}$

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 6. Recommended Operating Conditions

Symbol	Parameter	Conditions	EM74HC573			EM74HCT573			Unit
			Min	Typ	Max	Min	Typ	Max	
V_{CC}	supply voltage		2.0	5.0	6.0	4.5	5.0	5.5	V
V_I	input voltage		0		V_{CC}	0		V_{CC}	V
V_O	output voltage		0		V_{CC}	0		V_{CC}	V
T_{amb}	ambient temperature		-40	+25	+125	-40	+25	+125	$^{\circ}\text{C}$
$\Delta t/\Delta V$	input transition rise and fall rate	$V_{CC} = 2.0\text{ V}$			625				ns/V
		$V_{CC} = 4.5\text{ V}$		1.67	139		1.67	139	ns/V
		$V_{CC} = 6.0\text{ V}$			83				ns/V

9.Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HC573								
V _{IH}	HIGH-level input voltage	V _{CC} = 2.0 V	1.5			1.5		V
		V _{CC} = 4.5 V	3.15			3.15		V
		V _{CC} = 6.0 V	4.2			4.2		V
V _{IL}	LOW-level input voltage	V _{CC} = 2.0 V			0.5		0.5	V
		V _{CC} = 4.5 V			1.35		1.35	V
		V _{CC} = 6.0 V			1.8		1.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = -20μA; V _{CC} = 2.0 V	1.9			1.9		V
		I _O = -20μA; V _{CC} = 4.5 V	4.4			4.4		V
		I _O = -20μA; V _{CC} = 6.0 V	5.9			5.9		V
		I _O = -6.0 mA; V _{CC} = 4.5 V	3.84			3.7		V
		I _O = -7.8 mA; V _{CC} = 6.0 V	5.34			5.2		V
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = 20μA; V _{CC} = 2.0 V			0.1		0.1	V
		I _O = 20μA; V _{CC} = 4.5 V			0.1		0.1	V
		I _O = 20μA; V _{CC} = 6.0 V			0.1		0.1	V
		I _O = 6.0 mA; V _{CC} = 4.5 V			0.33		0.4	V
		I _O = 7.8 mA; V _{CC} = 6.0 V			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 6.0 V			±1		±1	μA
I _{OZ}	OFF-state output current	V _I = V _{IH} or V _{IL} ; V _{CC} = 6.0V ; V _O = V _{CC} or GND			±5		±10	μA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0A ; V _{CC} = 6.0 V			20		40	μA
C _I	input capacitance			3.5				pF

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HCT573								
V _{IH}	HIGH-level input voltage	V _{CC} = 4.5 V to 5.5 V	2.0			2.0		V
V _{IL}	LOW-level input voltage	V _{CC} = 4.5 V to 5.5 V			0.8		0.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = -20µA; V _{CC} = 4.5 V	4.4		4.4		V	
		I _O = -6.0 mA; V _{CC} = 4.5 V	3.84		3.7		V	
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = 20µA; V _{CC} = 4.5 V			0.1		0.1	V
		I _O = 6.0 mA; V _{CC} = 4.5 V			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 5.5 V			±1		±1	µA
I _{OZ}	OFF-state output current	V _I = V _{IH} or V _{IL} ; V _{CC} = 5.5 V ; V _O = V _{CC} or GND			±5		±10	µA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0 A ; V _{CC} = 5.5 V			20		40	µA
ΔI _{CC}	additional supply current	per pin ; V _I = V _{CC} - 2.1 V ; I _O = 0 A; other inputs at V _{CC} or GND; V _{CC} = 4.5 V to 5.5 V			450		490	µA
C _i	input capacitance			3.5				pF

[1] All typical values are measured at $T_{amb} = 25^\circ\text{C}$.

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Octal D-type transparent latch; 3-state

10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 10.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HC573								
t _{pd}	propagation delay	Dn to Qn; see Fig. 6 [2]						
		V _{CC} = 2.0 V			55		60	ns
		V _{CC} = 4.5 V			20		25	ns
		V _{CC} = 6.0 V			15		18	ns
		LE to Qn; see Fig. 7 [3]						
		V _{CC} = 2.0 V			55		60	ns
		V _{CC} = 4.5 V			20		25	ns
		V _{CC} = 6.0 V			15		18	ns
t _t	transition time	Qn; see Fig. 6						
		V _{CC} = 2.0 V			8		10	ns
		V _{CC} = 4.5 V			7		9	ns
		V _{CC} = 6.0 V			7		9	ns
t _{en}	enable time	$\overline{OE}$ to Qn; see Fig. 8 [3]						
		V _{CC} = 2.0 V			35		40	ns
		V _{CC} = 4.5 V			17		22	ns
		V _{CC} = 6.0 V			13		16	ns
t _{dis}	disable time	$\overline{OE}$ to Qn; see Fig. 8 [4]						
		V _{CC} = 2.0 V			37		42	ns
		V _{CC} = 4.5 V			17		22	ns
		V _{CC} = 6.0 V			13		15	ns
t _w	pulse width	LE HIGH; see Fig. 7						
		V _{CC} = 2.0 V	25			30		ns
		V _{CC} = 4.5 V	19			22		ns
		V _{CC} = 6.0 V	16			19		ns
t _{su}	set up time	Dn to LE; see Fig.9						
		V _{CC} = 2.0 V	16			20		ns
		V _{CC} = 4.5 V	13			15		ns
		V _{CC} = 6.0 V	11			13		ns
t _h	hold time	Dn to LE; see Fig.9						
		V _{CC} = 2.0 V	3			3		ns
		V _{CC} = 4.5 V	3			3		ns
		V _{CC} = 6.0 V	3			3		ns

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Octal D-type transparent latch; 3-state

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
C_{PD}	power dissipation capacitance	$f_i = 1 \text{ MHz};$ $V_i = \text{GND to } V_{CC};$ [6]		22				pF
EM74HCT573								
t_{pd}	propagation delay	Dn to Qn; see Fig. 6 [2]						
		$V_{CC} = 4.5 \text{ V}$			25		30	ns
		LE to Qn; see Fig. 7 [2]						
		$V_{CC} = 4.5 \text{ V}$			25		30	ns
t_t	transition time	Qn; see Fig. 6 [5]						
		$V_{CC} = 4.5 \text{ V}$			25		30	ns
t_{en}	enable time	$\overline{OE}$ to Qn; see Fig. 8 [3]						
		$V_{CC} = 4.5 \text{ V}$			20		25	ns
t_{dis}	disable time	$\overline{OE}$ to Qn; see Fig. 8 [4]						
		$V_{CC} = 4.5 \text{ V}$			20		25	ns
t_w	pulse width	LE HIGH; see Fig. 7						
		$V_{CC} = 4.5 \text{ V}$	19			22		ns
t_{su}	set up time	Dn to LE; see Fig.9						
		$V_{CC} = 4.5 \text{ V}$	13			15		ns
t_h	hold time	Dn to LE; see Fig.9						
		$V_{CC} = 4.5 \text{ V}$	3			3		ns
C_{PD}	power dissipation capacitance	$f_i = 1 \text{ MHz};$ $V_i = \text{GND to } V_{CC} - 1.5 \text{ V};$ [6]		22				pF

[1] Typical values are measured at $T_{amb} = 25^\circ\text{C}$.

[2] t_{pd} is the same as t_{PLH} and t_{PHL} .

[3] t_{en} is the same as t_{ON} and t_{PDS} .

[4] t_{dis} is the same as t_{PLZ} and t_{PHZ} .

[5] t_t is the same as t_{HL} and t_{LH} .

[6] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \sum (C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

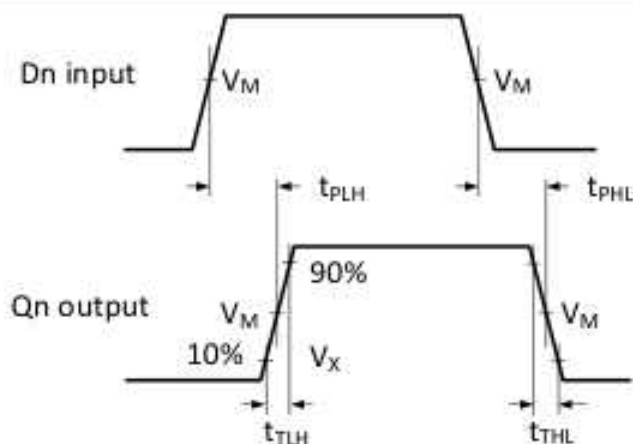
C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

N = number of inputs switching;

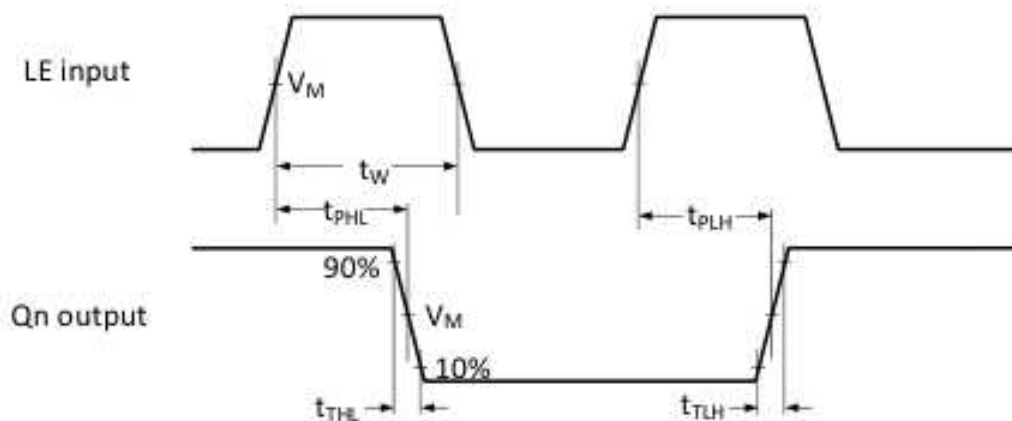
$\sum (C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

10.1. Waveforms and test circuit



Measurement points are given in Table 8.

Fig. 6. Propagation delay data input (Dn) to output (Qn) and output transition time

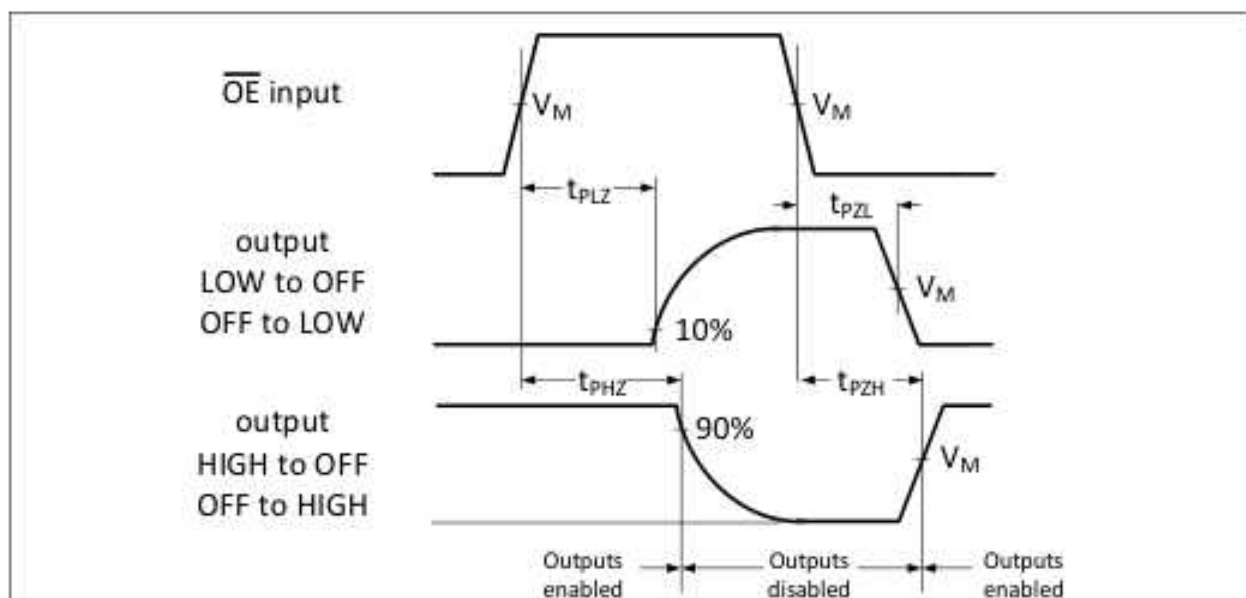


Measurement points are given in Table 8.

Fig. 7. Pulse width latch enable input (LE), propagation delay latch enable input (LE) to output (Qn) and output transition time

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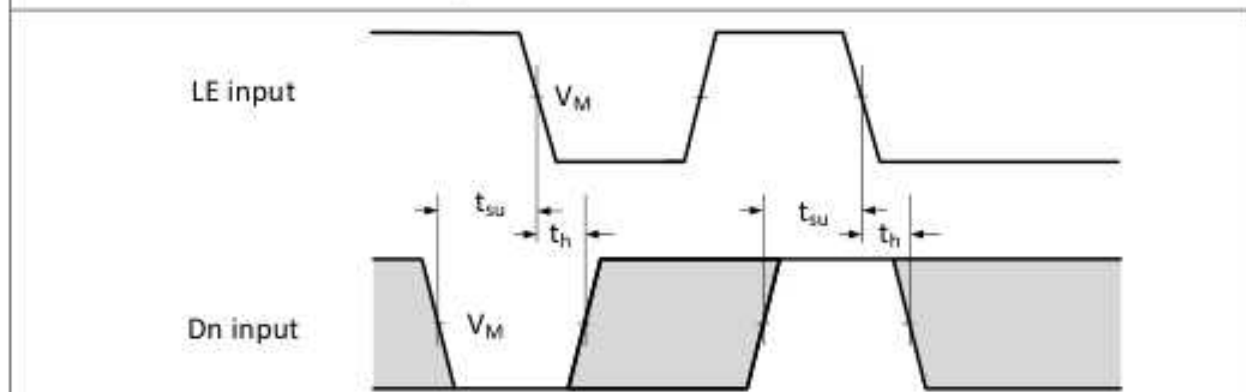
Octal D-type transparent latch; 3-state



Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 8. Enable and disable times



Measurement points are given in Table 8.

The shaded areas indicate when the input is permitted to change for predictable output performance.

Fig. 9. Set-up and hold times for data input (Dn) to latch input (LE)

Table 8. Measurement points

Type	Input	Output
	V_M	V_M
EM74HC573	0.5V _{CC}	0.5V _{CC}
EM74HCT573	1.3 V	1.3 V

EM74HC573; EM74HCT573

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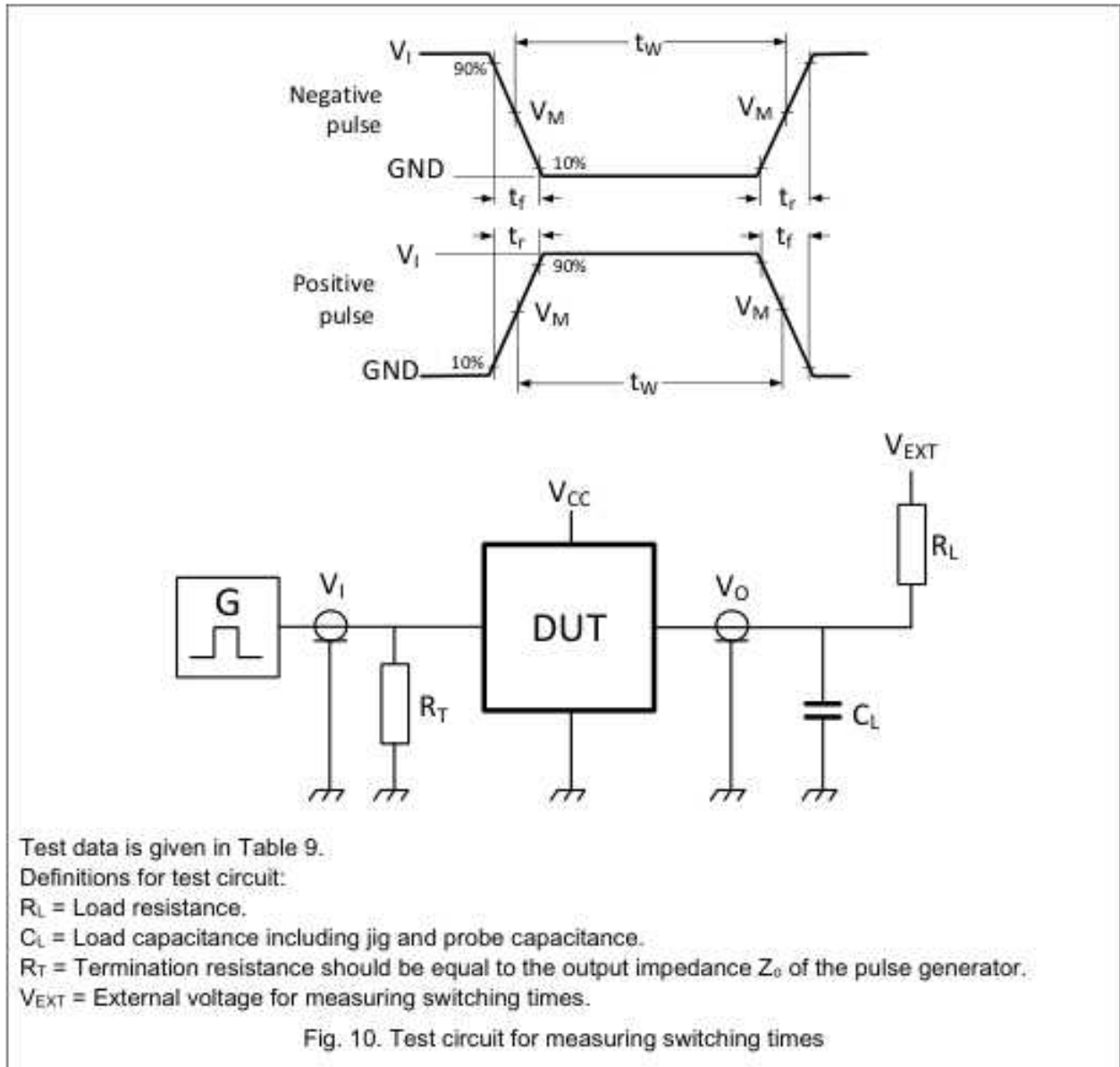
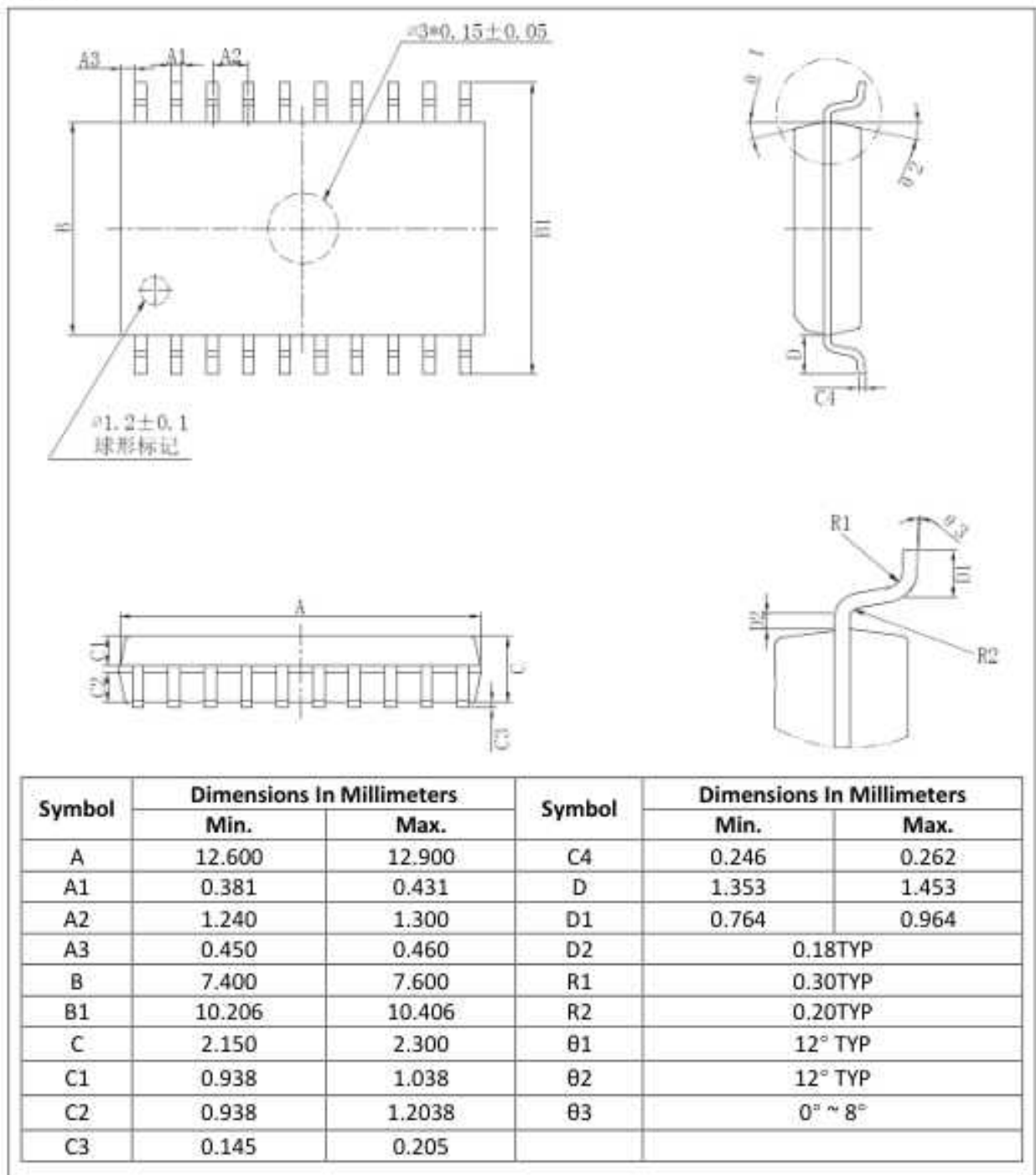


Table 9. Test data

Type	Input		Load		V_{EXT}		
	V_I	$t_r = t_f$	C_L	R_L	t_{PZL}, t_{PLZ}	t_{PHL}, t_{PLH}	t_{PZH}, t_{PHZ}
EM74HC573	V_{CC}	2.5 ns	15 pF	500 Ω	V_{CC}	Open	GND
EM74HCT573	3 V	2.5 ns	15pF	500 Ω	V_{CC}	Open	GND

11. Package Outline

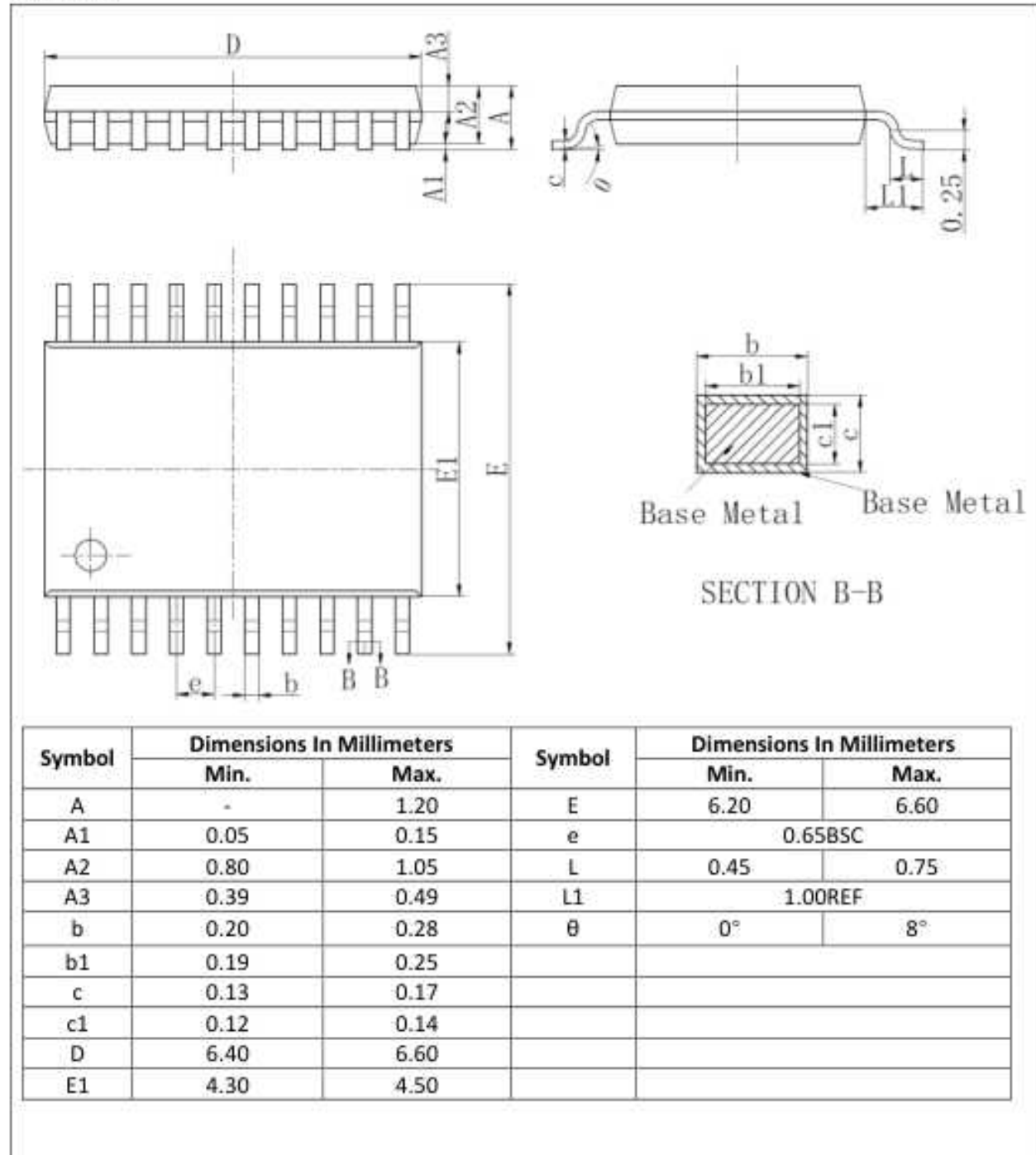
SOP-20L



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TSSOP-20L



12. Tape and Reel Information

12.1. Carrier tape dimensions

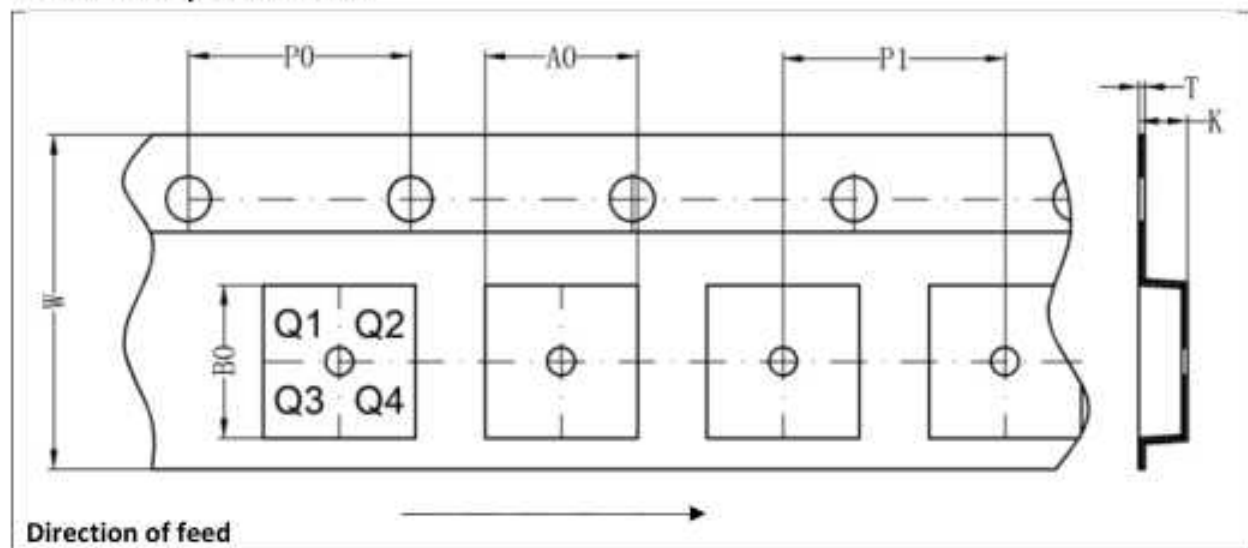
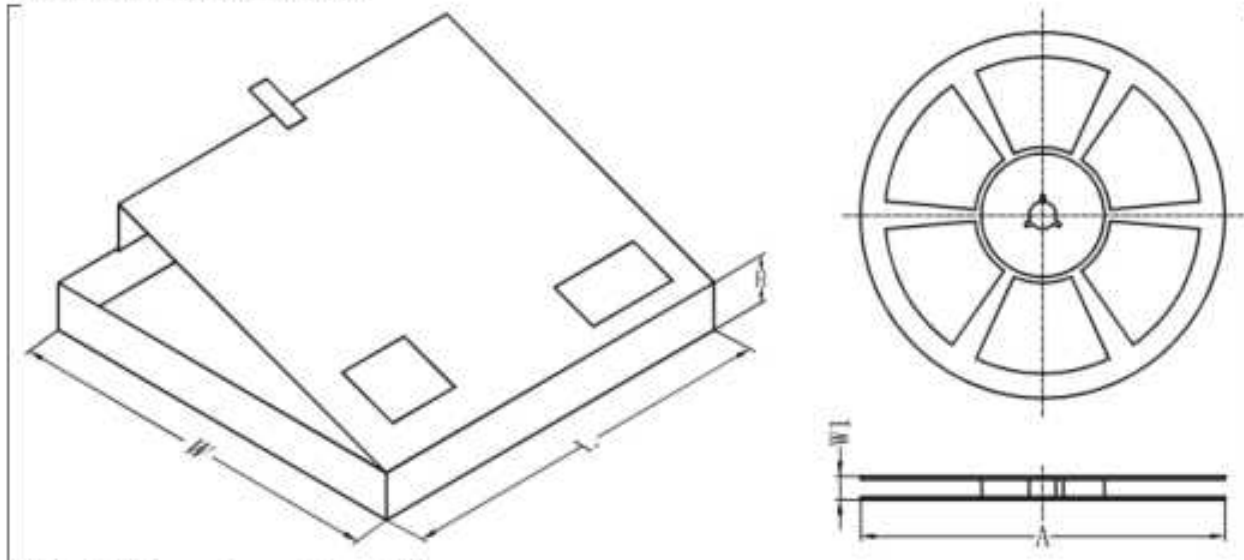


Table 10. Carrier tape dimensions

Package version	$A0$ (mm)	$B0$ (mm)	$K0$ (mm)	T (mm)	$P1$ (mm)	W (mm)	$P0$ (mm)	PIN 1
SOP-20L	10.65	13.2	3.2	0.3	12	24	4	Q1
TSSOP-20L	6.85	6.85	1.7	0.22	8	12	4	Q1

12.2. Reel and box dimensions

Table 11. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm) [2]
SOP-20L	D	330	30.4	2000	1	358x340x50
TSSOP-20L	PW	330	18.4	3000	1	358x340x50

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

13. Abbreviations

Table 12. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model
TTL	Transistor-Transistor Logic

14. Revision History

Table 13. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74HC_HCT573 Rev. 1.0	May 20, 2025	Product datasheet		